

HONOS

Enterprise Attendance System

Comprehensive User Guide & Technical Manual



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1. Core System Architecture

The Honos system uses a highly secure, offline-first data pipeline to ensure unbreakable biometric security combined with seamless cloud synchronization.

Attendance Verification Flow

1. **Initiation:** Supervisor initiates Check-In from the dashboard.
2. **Geofence Check:** System verifies current GPS coordinates are within the site radius.
3. **Liveness Detection:** AI engine prompts the guard to perform a randomized facial action (e.g., Smile, Blink) to prevent photo spoofing.
4. **Biometric Comparison:** Facial vectors are extracted and compared against the registered database entirely on-device.
5. **Synchronization:** Record is saved offline and instantly synced to Firebase Cloud Firestore when internet is available.

2. Administrator Guide

The Administrator has global oversight over the entire system. Login with the **Admin Credentials** to access the Command Center.

2.1 Manage Supervisors

Supervisors are the field operators responsible for managing specific sites and guards.

- 1 From the Admin Dashboard, tap on the **Manage Supervisors** action card.
- 2 Tap the **+** (**Add**) button in the bottom right corner.
- 3 Fill out the Supervisor's Name, Username, and secure Password.
- 4 Tap **Save**. The supervisor can now log into the app using these credentials.
- 5 To delete a supervisor, swipe left on their name in the list and tap the delete icon.

2.2 Manage Sites & Geofencing

Sites dictate where attendance is allowed. A strict Geofence radius ensures guards are physically present.

- 1 From the Admin Dashboard, tap on **Sites & Geofencing**.
- 2 Tap the **+** (**Add**) button to open the interactive map.
- 3 **Drop the Pin:** Drag the map to place the red marker exactly over your facility's location.
- 4 **Define Perimeter:** Enter a radius in meters (e.g., 200m). A green circle will visually confirm the geofenced area.
- 5 Assign a registered Supervisor to this site from the dropdown list.
- 6 Tap **Save Site**.

Note: Guards assigned to this site will strictly be unable to check-in if their phone's GPS is outside the defined radius.

2.3 Manage Guards & Staff

This is where you register the workforce and map their facial biometric data into the system.

- 1 Navigate to **Guards & Staff** from the Command Center.
- 2 Tap the **+** (**Add**) button.
- 3 Enter the guard's demographic details (Name, Employee ID, Phone Number, Salary, Join Date).
- 4 Assign the guard to a specific **Site** and **Supervisor**.
- 5 **Crucial Step - Face Registration:** Tap the "Register Face" button. Position the guard's face within the camera frame. The system will mathematically map their facial structure and bind it to their profile.
- 6 Tap **Save Guard**.

2.4 Supervisor Live Tracker

Track the real-time geographic locations of all active supervisors.

- 1 Tap on **Live Supervisor Map**.
- 2 The map will populate with markers representing your supervisors.
- 3 The app runs a persistent background service on the Supervisor's phone, syncing their location to the cloud every few minutes.

2.5 Attendance Reports & Analytics

View, filter, and export the comprehensive attendance logs generated by the field supervisors.

- 1 Navigate to **Attendance Report**.
- 2 You will see a chronologically sorted list of all attendance events (Check-Ins and Check-Outs).
- 3 Each log displays the Guard's Name, Timestamp, Exact GPS Location at the time of scanning, and Status (Present/Late).
- 4 As an Admin, you possess the exclusive authority to delete erroneous records by tapping the delete icon on a specific log, or clear the entire history via the top-right menu.

3. Supervisor Guide

The Supervisor dashboard is optimized for rapid, on-the-field attendance processing.

3.1 Dashboard Overview

Upon login, Supervisors instantly see a summary of their assigned Site, total assigned guards, and the number of guards currently checked in today.

3.2 Marking Guard Attendance (Check-In / Out)

This is the core operational flow for the Supervisor.

- 1 From the Supervisor Dashboard, tap **Mark Attendance**.
- 2 **Select Guard:** You will see a list of guards assigned to your specific site. Tap on a guard.
- 3 **Choose Action:** Select the **Green Check-In** or **Red Check-Out** button.
- 4 **Liveness Verification:** The camera will open. The guard must perform the random AI challenge displayed on screen (e.g., "Blink", "Smile"). This prevents spoofing via photos.
- 5 **Biometric Match:** Once liveness is verified, the system scans the face and compares it to the Guard's registered profile vector.
- 6 **Geofence Validation:** Simultaneously, the app ensures your current GPS coordinates fall within the Site's radius.
- 7 **Success:** If all checks pass, the attendance record is permanently logged to the offline database and immediately synced to the Admin Cloud.

The first part of the paper discusses the importance of the research and the objectives of the study. It then presents a literature review of the existing research on the topic. The second part of the paper describes the methodology used in the study, including the data collection and analysis techniques. The third part of the paper presents the results of the study, and the fourth part discusses the implications of the findings.

The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables.

The results of the study show that there is a significant positive relationship between the variables. This finding is consistent with the previous research in the field. The implications of the findings suggest that the research has practical applications in the field.

In conclusion, the study has shown that there is a significant positive relationship between the variables. This finding is consistent with the previous research in the field. The implications of the findings suggest that the research has practical applications in the field.

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